InvoiceAI — Machine Learning Canvas

Obligation-readiness triage for Accounts Payable · scoping & planning view

Value proposition · Predict · Learn · Monitoring

PREDICTION TASK

- Multi-stage: object detection (bbox + class per region, 14 region classes + stamp/signature) → OCR on detected crops → rule-based verdict.
- Entity: one invoice image. Outcome: Ready / Needs review / Not ready, observed.

DECISIONS

- Verdict routes the invoice: auto-accept as a structured obligation record, or flag for human review naming the missing field(s).
- Policy rules (visual mark, reference no., date range, payment terms) are user-configurable and fail-closed.

VALUE PROPOSITION

- For back-office/AP teams and platforms like Pistac.io turning scanned invoices into structured obligation records.
- Replaces manual, inconsistent per-invoice checks for stamps, signatures, reference numbers and terms.
- Now live as a callable FastAPI service (port 8000) plus a lightweight browser front-end — not just a demo UI.

DATA COLLECTION

- Initial set sourced from Kaggle invoice/OCR datasets plus a custom hand-annotated subset (50–150 imgs) for labels absent from public data.
- At inference, the raw input is a client-uploaded invoice image; nothing is retained beyond the request.

DATA SOURCES

- Kaggle: High-Quality Invoice Images for OCR; OCR Dataset of Multi-type Documents.
- Kaggle: SignverOD (signatures), StaVer (stamps) — kept as two independent detection classes.

IMPACT SIMULATION

- Cost of a missed field: an invoice wrongly marked Ready skips human review — the fail-closed default is the mitigation.
- Deployment criteria: /health confirms both detector weight sets loaded before traffic is trusted.

MAKING PREDICTIONS

- Real-time, one invoice at a time, via POST /predict (multipart upload).
- ~15–20s per invoice, CPU-only (no GPU in production) on a GCP e2-medium VM.

BUILDING MODELS

- 3 models in production: region detector (YOLOv8n, 14 classes), stamp/signature detector (YOLOv8n, 2 classes), EasyOCR (off-the-shelf).
- SSD was taught; YOLOv8n is the trained artifact — a documented substitution given the project timeline.

FEATURES

- Preprocessed image pixels (grayscale, resize, denoise, deskew) feed the detectors directly — no hand-crafted features.
- OCR text is parsed with keyword/regex matching for 6 reference-number types.

MONITORING

- Live-verified: /health and /policies return correct, schema-matching JSON; a blank test invoice correctly triggers fail-closed "Not ready" on every policy preset.
- docker logs on the VM is the primary signal today — no automated alerting yet, a stated limitation for this graded deployment.
- Known issue under review: the image bundles unused CUDA libraries despite CPU-only inference — caused a real disk-space failure, resolved by resizing 30GB → 100GB.